

The Carbon Remix: Scaling Carbon Insights with an AI Collaborator

Overview

In this module, you will investigate how ecosystems "breathe." You will start with a trusted scientific tutorial from the **National Ecological Observatory Network (NEON)** to visualize carbon exchange. Then, you will act as a "Code Editor," using an AI assistant to modify that code to answer a research question of your own.

The Goal: Move from being a passive user of a tutorial to an active investigator using AI as a technical partner.

Part A: FOUNDATIONS

Background: The "Breath" of the Forest

Terrestrial ecosystems play a massive role in the global carbon cycle. Through **photosynthesis**, plants pull carbon dioxide (CO₂) out of the atmosphere. Through **respiration**, plants and soil microbes release CO₂ back into the air.

Scientists measure this exchange using a method called **Eddy Covariance**. Imagine standing on a tower above a forest: sensors measure the vertical "eddies" (swirls) of wind. By measuring the CO₂ concentration in these upward and downward gusts, we can calculate the **Net Ecosystem Exchange (NEE)**—the total balance of carbon moving in or out.

Key Terms to Know:

- **Flux:** The rate of transfer of a fluid or gas (like CO₂) across a surface over time.
- **NEON:** A continental-scale observation system that collects long-term data on the impacts of climate change.
- **CSV File:** A "Comma Separated Values" file (basically a simple spreadsheet) where data is stored.
- **Roo:** An AI "Agent" that lives inside your code editor (VS Code) and can read, explain, and help you edit your data analysis scripts.

Software Setup

Before we begin, ensure you have the following ready:

1. **VS Code:** Our primary coding environment.
2. **Roo Extension:** Click the "Extensions" icon in VS Code (looks like four squares) and ensure "Roo" is installed and active.
3. **The Tutorial File:** Open the file named NEON_CarbonFlux_Intro.ipynb. This is a **Jupyter Notebook**, which allows us to mix text and code.

Part B: Guided Inquiry & Analysis

Goal: Understand the data using code in collaboration with an AI assistant.

1. **Open the Notebook:** In VS Code, open the NEON_CarbonFlux_Intro.ipynb file.
2. **Run the Code:** Click "Run All" at the top of the notebook.
 - *Note:* It may take a minute to download the NEON data from the cloud.
3. **Inspect the Output:** Look at the plot generated at the bottom. It likely shows \$CO_2\$ flux over a 24-hour period.
 - **Question:** Why does the \$CO_2\$ flux drop below zero during the day? (*Hint: What are the plants doing when the sun is out?*)
4. **AI Checkpoint (Explanation):** Highlight a line of code you don't understand (e.g., something involving pandas or matplotlib).
 - Open the Roo sidebar and type: *"Roo, I'm an environmental science student. Can you explain what this specific line of code is doing in plain English?"*
 - **Action:** Paste Roo's explanation as a comment in your notebook using the # symbol.

Goal: Use Roo to help you pivot the analysis toward a new question.

Now that you have working code, you are going to change it. Instead of just looking at the daily cycle, we want to see how **Environmental Variables** impact that cycle.

Step 1: Choose Your Modification

Select **one** of the following "New Questions" to investigate:

- **The Temperature Factor:** How does \$CO_2\$ flux change as air temperature increases?
- **The Light Factor:** How does "Photosynthetically Active Radiation" (PAR) correlate with the rate of \$CO_2\$ uptake?
- **The Storm Factor:** What happens to the carbon flux data during a high-wind event or a heavy rain day?

Step 2: Prompting Roo for Changes

Don't ask Roo to "Write a new script." Instead, ask it to **modify** the existing tutorial.

- **Example Prompt:** *"Roo, I am using the NEON flux tutorial. I want to modify the final plot. Instead of plotting \$CO_2\$ flux vs. Time, can you help me create a scatter plot that shows \$CO_2\$ flux on the y-axis and Air Temperature on the x-axis?"*

Step 3: Human-in-the-Loop Checkpoint (Critical Review)

When Roo provides code, **do not just copy-paste**. Review it first:

1. **Logical Check:** Does the new code still use the same data source (the NEON dataframe)?
2. **Scientific Check:** Run the modified code. If Roo gives you a plot where photosynthesis is happening at 2:00 AM (midnight), the AI has "hallucinated" or made a logic error.
3. **The "Fix":** If the results look physically impossible, tell Roo: *"That doesn't look right biologically because photosynthesis requires light. Can you check if the time-stamping is correct?"*

Part C: Synthesis & Communication

Goal: Document your scientific process and the role of the AI.

In a separate document (or a new text cell in your notebook), provide the following:

1. **The Result:** A screenshot of your **New Modified Plot**.
2. **The Interpretation:** In 2–3 sentences, what does your new plot tell you about the ecosystem? (e.g., "As temperature increased beyond 30°C, we saw the \$CO_2\$ uptake begin to slow down, likely due to plant heat stress.")
3. **The AI Reflection (The Token Ledger):**
 - How many prompts did you give Roo?
 - Name one time Roo gave you a suggestion that you had to **correct** or **reject**.
 - How did having an AI partner change how you felt about working with complex scientific data?